Etiology and Pathogenesis

Nearly all patients with diabetes have cutaneous findings related to their condition. Some of these skin manifestations are a direct result of metabolic changes such as hyperglycemia and hyperlipidemia. Progressive damage to the vascular, neurologic, and immune systems also plays a significant role in the development of skin disorders. The underlying mechanisms for certain diabetes-associated skin conditions remain unclear.

Hyperglycemia leads to nonenzymatic glycosylation (NEG) of structural and regulatory proteins, including collagen. Although NEG occurs naturally during aging, it is markedly accelerated in diabetes.[6,7] This process results in the formation of advanced glycation end products (AGEs), which reduce the acid solubility and enzymatic digestibility of cutaneous collagen. Disorders such as **diabetic thick skin** and **limited joint mobility (LJM)** are believed to result directly from the accumulation of AGEs. Research indicates that levels of cutaneous AGEs strongly correlate with microvascular complications of diabetes, including retinopathy and nephropathy.

Immune dysfunction is also common in diabetes. Hyperglycemia and ketoacidosis impair white blood cell functions such as chemotaxis, phagocytosis, and bactericidal activity.[10] While infections were once a leading cause of death in diabetic patients, improved glycemic control and antibiotic therapy have significantly reduced mortality. Nevertheless, certain severe infections—such as **malignant external otitis**, **necrotizing fasciitis**, and **mucormycosis**—occur more frequently in individuals with diabetes.[11]

Metabolic abnormalities, particularly hyperinsulinemia seen in early insulin-resistant type 2 diabetes, contribute to specific skin changes. Insulin's interaction with the insulin-like growth factor-1 (IGF-1) receptor is thought to drive abnormal epidermal proliferation, leading to **acanthosis nigricans**.[12] Additionally, insulin deficiency disrupts lipid metabolism, as insulin regulates lipoprotein lipase (LPL) activity.[13] Without sufficient insulin, triglyceride-rich lipoproteins are not properly metabolized, resulting in severe hypertriglyceridemia and the development of **eruptive xanthomas**. Dyslipidemia also contributes to the macro- and microangiopathies associated with diabetes.

Vascular abnormalities further exacerbate cutaneous complications. Diabetic patients exhibit increased vascular permeability, impaired sympathetic vascular responsiveness, and reduced adaptation to thermal and hypoxic stress.[10] These microvascular defects, combined with large-vessel arteriosclerosis, predispose individuals to poor wound healing and the formation of **diabetic foot ulcers**.

Cutaneous Findings in Diabetes

- **Acanthosis nigricans**
- **Limited joint mobility** and **scleroderma-like syndrome**
- **Scleredema diabeticorum**
- **Eruptive xanthomas**
- **Bacterial infections** (e.g., streptococcal, malignant external otitis, necrotizing fasciitis)
- **Fungal infections** (e.g., candidiasis, onychomycosis, mucormycosis)
- **Foot ulcers**
- **Necrobiosis lipoidica**
- **Granuloma annulare**
- **Diabetic dermopathy**
- **Acquired perforating disorders**
- **Bullosis diabeticorum**

Related Systemic Features

- Retinopathy
- Nephropathy
- Neuropathy
- Cardiovascular disease

- Peripheral vascular disease
- Hyperlipidemia
- Hypertension

Diagnosis

- In asymptomatic individuals, diagnosis requires two abnormal tests:
- Fasting blood glucose ≥ 126 mg/dL, or
- Random blood glucose ≥ 200 mg/dL.
- **Hemoglobin A1C** is increasingly used alongside fasting plasma glucose for diagnosis.

Management

- **Diet and exercise** as foundational therapy.
- Referral to a primary care provider or endocrinologist for initiation of **oral hypoglycemic agents** or **insulin**.
- **Routine foot examination** in all diabetic patients.
- Address **modifiable risk factors** (e.g., smoking, hypertension, dyslipidemia).